Motion-First Geometry, Lorentz Ether vs. Special Relativity, Poincaré's Fuchsian/Lorentz Group Theory, and Casimir/Wigner $\mathrm{ISO}(1,3)$ Classification

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Abstract

This treatise synthesizes four foundational investigations authored in May 2026 during the development of the Frohmanian Symplectic Tether program: (I) *Motion-First Geometry and Relational Dynamics*, contrasting relational kinematic axioms (Heraclitus, Leibniz, Barbour–Bertotti shape space, and Rovelli relational quantum mechanics) against static substantivalist spacetime blocks; (II) *Lorentz Ether Theory (LET) vs. Einstein Special Relativity (SR)*, analyzing empirical equivalence under the Lorentz transformations alongside ontological divergence between dynamical stress in a preferred frame and Minkowski metric kinematics; (III) *Poincaré's Group-Theoretic Synthesis (1880–1906)*, tracing the mathematical continuity from Fuchsian automorphic groups and the hyperbolic disk model of $\mathrm{PSL}(2,\mathbb{R}) \cong \mathrm{SO}^+(1,2)$ to the 1905/1906 Palermo paper establishing the Lorentz group $\mathrm{SO}^+(1,3)$, its four-vector invariants, and the 10-dimensional inhomogeneous Poincaré group $\mathrm{ISO}(1,3) = \mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3)$; and (IV) *Casimir Invariants and Wigner's 1939 Unitary Irreducible Representation Classification*, deriving how the two polynomial Casimir operators $C_1 = P_\mu P^\mu$ and $C_2 = W_\mu W^\mu$ (with Pauli–Lubanski pseudovector $W^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} J_{\nu\rho} P_\sigma$) classify elementary relativistic particles solely by invariant mass m^2 and spin/helicity (s, λ) across the massive ($m^2 > 0$, SU(2) little group), massless ($m^2 = 0$, ISO(2) little group), and tachyonic ($m^2 < 0$, SO(1,2) little group) orbits.

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1 Provenance and Archival Record

- **Author:** Benjamin Stanley Frohman (`ORCID: 0009-0006-7068-3718`).
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- **Repository Companion Files:** Includes both the verbatim archival RTF/plaintext transcript (`Grok_Validation_Proof_2026-05-19.rtf`) and this typeset mathematical monograph.

2 Part I: Motion-First Physics and Relational Shape Dynamics

Standard textbook presentations of mechanics and relativity begin by postulating a fixed background manifold $\mathcal{M}$ equipped with a metric tensor $g_{\mu\nu}$ and subsequently define motion as a parameterized curve $\gamma : \mathbb{R} \to \mathcal{M}$. By contrast, a *motion-first* ontology takes relational flux and state transitions as primitive, recovering spatial geometry and temporal duration as emergent invariants of the relational configuration space.

2.1 Shape Space and Temporal Emergence

Let $\mathcal{Q} = (\mathbb{R}^3)^N$ denote the configuration space of N particles, and let $\mathrm{Sim}(3) = \mathbb{R}^3 \rtimes (\mathrm{SO}(3) \times \mathbb{R}_+)$ denote the similarity group of translations, rotations, and global dilatations. In relational mechanics (Leibniz–Mach–Barbour–Bertotti), the physical configuration space is the quotient *shape space*

$$\mathcal{S} = \mathcal{Q}/\mathrm{Sim}(3).$$

Rather than evolving with respect to an external Newtonian clock t, trajectories correspond to geodesics on $\mathcal{S}$ under a Jacobi-type Maupertuis action

$$S_{\mathrm{rel}} = \int \sqrt{2(E - V(q))\, T_{\mathrm{best-matched}}(\dot{q}, \dot{q})}\, d\lambda,$$

where the ephemeral time increment dt_{eph} is defined directly from the relational kinetic change itself:

$$dt_{\mathrm{eph}} = \sqrt{\frac{T_{\mathrm{best-matched}}(dq, dq)}{2(E - V(q))}}.$$

Thus time is not an external container in which motion occurs; time is the canonical measure of relational change.

3 Part II: Lorentz Ether Theory vs. Einstein's Special Relativity

Both Lorentz–Poincaré Ether Theory (LET, 1892–1905) and Einstein's Special Relativity (SR, 1905) share the exact same kinematic coordinate transformation law between inertial frames moving at relative velocity v along the x-axis:

$$t' = \gamma\Big(t - \frac{vx}{c^2}\Big), \qquad x' = \gamma(x - vt), \qquad y' = y, \qquad z' = z, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Consequently, both frameworks yield identical predictions for all Michelson–Morley, Kennedy–Thorndike, Ives–Stilwell, and synchrotron experiments. Their divergence is structural and geometric:

Feature	Lorentz Ether Theory (LET)	Einstein Special Relativity (SR)
Ontological Primitive	Preferred rest frame Σ_0 (ether medium) with absolute simultaneity	No preferred inertial frame; Minkowski spacetime $(\mathbb{R}^{1,3}, \eta_{\mu\nu})$
Length Contraction	Dynamical electromagnetic compression of intermolecular forces moving through Σ_0	Reciprocal geometric projection of worldtubes across non-orthogonal simultaneity hyperplanes
Time Dilation	Dynamical slowing of physical oscillators ("local time" $t' = t - vx/c^2$ upgraded to exact γ-law)	Proper time $\Delta\tau = \int \sqrt{-ds^2}/c$ along timelike curves
Group Symmetry	Observational symmetry masking an asymmetric substrate	Fundamental spacetime symmetry group $\mathrm{ISO}(1,3)$

4 Part III: Poincaré's Group-Theoretic Arc (1880–1906)

Henri Poincaré's 1905 Comptes Rendus note and 1906 Rendiconti del Circolo Matematico di Palermo memoir (*Sur la dynamique de l'électron*) were the culmination of a 25-year program applying Lie's theory of continuous transformation groups to geometry and mathematical physics:

1. **Fuchsian Groups and Hyperbolic Geometry (1880–1882):** Poincaré recognized that discrete Fuchsian groups of fractional linear transformations $z \mapsto \frac{az+b}{cz+d}$ preserving the unit disk $\mathbb{D}$ are isomorphic to isometries of the hyperbolic plane, with $\mathrm{PSL}(2,\mathbb{R}) \cong \mathrm{SO}^+(1,2)$ preserving the ternary quadratic form $x_0^2 - x_1^2 - x_2^2$.

2. **Foundations of Geometry via Lie Groups (1887–1898):** In *On the Fundamental Hypotheses of Geometry* (1887) and *Science and Hypothesis* (1902), Poincaré demonstrated that homogeneous two-dimensional Riemannian geometries of constant curvature correspond bijectively to three-parameter Lie groups ($\mathrm{SO}(3)$, $\mathrm{ISO}(2)$, $\mathrm{SO}^+(1,2)$), formulating the thesis that geometry is the invariant theory of a transformation group arising from physical displacements.

3. **The Lorentz Group and Four-Vector Formalism (1905–1906):** Poincaré proved that the transformations preserving Maxwell's equations form a Lie group (which he named the *Lorentz group* $\mathrm{O}(1,3)$), introduced the imaginary time coordinate $x_4 = ict$ so that the interval

$$s^2 = x^2 + y^2 + z^2 - c^2t^2 = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

   is preserved as a four-dimensional orthogonal rotation, constructed four-vectors for position, velocity, current density, and force, and identified the infinitesimal generators of boosts K_i and rotations J_i.

5 Part IV: Casimir Invariants and Wigner's $\mathrm{ISO}(1,3)$ Classification

The inhomogeneous Lorentz (Poincaré) group $\mathcal{P} = \mathrm{ISO}^+(1,3) = \mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3)$ is a 10-dimensional non-compact Lie group generated by four spacetime translations P^μ and six antisymmetric Lorentz

generators $J^{\mu\nu} = -J^{\nu\mu}$ (with $J^i = \frac{1}{2}\varepsilon^{ijk}J_{jk}$ and $K^i = J^{0i}$) satisfying the Lie algebra $\mathfrak{iso}(1,3)$:

$$[P^\mu, P^\nu] = 0, \tag{1}$$
$$[J^{\mu\nu}, P^\rho] = i\hbar\big(\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu\big), \tag{2}$$
$$[J^{\mu\nu}, J^{\rho\sigma}] = i\hbar\big(\eta^{\nu\rho}J^{\mu\sigma} - \eta^{\mu\rho}J^{\nu\sigma} - \eta^{\nu\sigma}J^{\mu\rho} + \eta^{\mu\sigma}J^{\nu\rho}\big). \tag{3}$$

Theorem 5.1 (Casimir Invariants and Wigner's Little-Group Classification (1939))**.** *Let $W^\mu := \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}J_{\nu\rho}P_\sigma$ denote the Pauli–Lubanski pseudovector, which satisfies $P_\mu W^\mu = 0$ and $[P^\mu, W^\nu] = 0$. The center $\mathcal{Z}(\mathcal{U}(\mathfrak{iso}(1,3)))$ of the universal enveloping algebra is generated by the two quadratic and quartic Casimir operators:*

$$C_1 = P_\mu P^\mu, \qquad C_2 = W_\mu W^\mu.$$

On any unitary irreducible representation $\mathcal{H}$ of the universal cover $\mathbb{R}^{1,3} \rtimes \mathrm{SL}(2,\mathbb{C})$ (by Schur's Lemma):

1. ***Massive Class** ($C_1 = m^2c^2 > 0$, $P^0 > 0$): In the rest frame $p^\mu = (mc, \mathbf{0})$, the stabilizer (little group) is $\mathrm{SU}(2)$ (double cover of $\mathrm{SO}(3)$), $W^\mu = (0, mc\mathbf{S})$, and*

   $$C_1 = m^2c^2, \qquad C_2 = -m^2c^2\hbar^2 s(s+1), \qquad s \in \big\{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, 2, \dots\big\},$$

   yielding $2s+1$ polarization states $s_z \in \{-s, \dots, +s\}$.

2. ***Massless Finite-Spin Class** ($C_1 = 0$, $C_2 = 0$, $P^\mu \neq 0$): In the standard lightlike frame $p^\mu = (E/c, 0, 0, E/c)$, the little group is the Euclidean group $\mathrm{ISO}(2) = \mathbb{R}^2 \rtimes \mathrm{SO}(2)$ (double cover). When the non-compact "translational" generators Π_1, Π_2 act trivially, $W^\mu = \lambda P^\mu$ where the helicity $\lambda = \frac{\mathbf{J}\cdot\mathbf{P}}{|\mathbf{P}|} \in \{0, \pm\frac{1}{2}, \pm1, \pm\frac{3}{2}, \pm2, \dots\}$ is Lorentz-invariant.*

3. ***Massless Continuous-Spin Class** ($C_1 = 0$, $C_2 = -\rho^2 < 0$): Non-trivial action of $\mathbb{R}^2 \subset \mathrm{ISO}(2)$ yields infinite-dimensional helicity towers.*

4. ***Tachyonic Spacelike Class** ($C_1 = -\mu^2c^2 < 0$): Little group $\mathrm{SO}^+(1,2) \cong \mathrm{PSL}(2,\mathbb{R})$, whose non-trivial unitary representations are infinite-dimensional.*

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